# Model 1 — Classical Machine Learning Baseline (Random Forest + XAI)

This section establishes a classical benchmark for intrusion/anomaly detection using the NSL-KDD dataset. A Random Forest classifier is used as a strong, widely adopted baseline for structured cybersecurity data. The pipeline applies one-hot encoding for categorical features and feature scaling for consistency with later quantum and hybrid models.

Performance is evaluated using Accuracy, ROC-AUC, and PR-AUC (important for imbalanced detection tasks), along with ROC/PR curves and a confusion matrix. Explainable AI is integrated using SHAP to provide both global and local interpretability.

```
# core ML + plotting + XAI
!pip -q install pandas numpy scikit-learn matplotlib seaborn shap j
```

```
import os, zipfile
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, classification_report, confusion_matrix,
    roc_auc_score, average_precision_score,
    RocCurveDisplay, PrecisionRecallDisplay
)

import shap
import joblib
```

```
# Locate KDDTrain+.txt and KDDTest+.txt automatically so the notebc
# if the zip contains extra folders.

ZIP_PATH = "archive-2.zip"
EXTRACT_DIR = "nslkdd_data"

if not os.path.exists(ZIP_PATH):
    raise FileNotFoundError(f"Upload '{ZIP_PATH}' in Colab -> Files

os.makedirs(EXTRACT_DIR, exist_ok=True)

with zipfile.ZipFile(ZIP_PATH, "r") as z:
    z.extractall(EXTRACT_DIR)

train_file, test_file = None, None
for root, _, files in os.walk(EXTRACT_DIR):
    for f in files:
        if f == "KDDTrain+.txt":
            train_file = os.path.join(root, f)
        if f == "KDDTest+.txt":
            test_file = os.path.join(root, f)

if not train_file or not test_file:
    raise FileNotFoundError("Could not find KDDTrain+.txt or KDDTes

print("Train:", train_file)
print("Test :", test_file)
```

```
Train: nslkdd_data/archive-2/nsl-kdd/KDDTrain+.txt
Test : nslkdd_data/archive-2/nsl-kdd/KDDTest+.txt
```

```
# NSL-KDD text files do not include headers; so define them explici

COLUMNS = [
    "duration","protocol_type","service","flag","src_bytes","dst_by
    "wrong_fragment","urgent","hot","num_failed_logins","logged_in"
    "root_shell","su_attempted","num_root","num_file_creations","nu
    "num_access_files","num_outbound_cmds","is_host_login","is_gues
    "srv_count","serror_rate","srv_serror_rate","rerror_rate","srv_
    "same_srv_rate","diff_srv_rate","srv_diff_host_rate","dst_host_
    "dst_host_srv_count","dst_host_same_srv_rate","dst_host_diff_sr
    "dst_host_same_src_port_rate","dst_host_srv_diff_host_rate","ds
    "dst_host_srv_serror_rate","dst_host_rerror_rate","dst_host_srv
    "label","difficulty"
]

train_df = pd.read_csv(train_file, names=COLUMNS)
test_df  = pd.read_csv(test_file,  names=COLUMNS)

print("Train shape:", train_df.shape)
print("Test shape :", test_df.shape)
train_df.head(3)
```

```
Train shape: (125973, 43)
Test shape : (22544, 43)
```

| | duration | protocol_type | service | flag | src_bytes | dst_bytes | land |
|---|---|---|---|---|---|---|---|
| **0** | 0 | tcp | ftp_data | SF | 491 | 0 | 0 |
| **1** | 0 | udp | other | SF | 146 | 0 | 0 |
| **2** | 0 | tcp | private | S0 | 0 | 0 | 0 |

3 rows × 43 columns

```
# For a strong baseline and fair comparison across Classical vs Qua
# Using binary labels: normal (0) vs attack (1).

train_df["label"] = train_df["label"].apply(lambda x: 0 if x == "no
test_df["label"]  = test_df["label"].apply(lambda x: 0 if x == "nor

# 'difficulty' is metadata provided by NSL-KDD, not a traffic featu
train_df.drop(columns=["difficulty"], inplace=True)
test_df.drop(columns=["difficulty"], inplace=True)

print("Train class counts:\n", train_df["label"].value_counts())
print("\nTest class counts:\n", test_df["label"].value_counts())
```

```
Train class counts:
 label
0    67343
1    58630
Name: count, dtype: int64

Test class counts:
 label
1    12833
0     9711
Name: count, dtype: int64
```

```
# protocol_type, service, flag are categorical.
# one-hot encode and then align train/test columns to keep the feat

categorical_cols = ["protocol_type", "service", "flag"]

train_df = pd.get_dummies(train_df, columns=categorical_cols)
test_df  = pd.get_dummies(test_df,  columns=categorical_cols)

train_df, test_df = train_df.align(test_df, join="left", axis=1, fi

print("After encoding -> Train:", train_df.shape, "| Test:", test_d
```

```
After encoding -> Train: (125973, 123) | Test: (22544, 123)
```

```
X_train = train_df.drop(columns=["label"])
y_train = train_df["label"].astype(int)

X_test  = test_df.drop(columns=["label"])
y_test  = test_df["label"].astype(int)

scaler = StandardScaler()
X_train_sc = scaler.fit_transform(X_train)
X_test_sc  = scaler.transform(X_test)

print("Scaled matrices:", X_train_sc.shape, X_test_sc.shape)
```

```
Scaled matrices: (125973, 122) (22544, 122)
```

```
# Random Forest is used as the classical baseline because it perfor
# is fast/reproducible, and integrates well with SHAP for explainab

rf = RandomForestClassifier(
    n_estimators=200,
    random_state=42,
    n_jobs=-1,
    class_weight="balanced"
)

rf.fit(X_train_sc, y_train)
print("Random Forest trained")
```

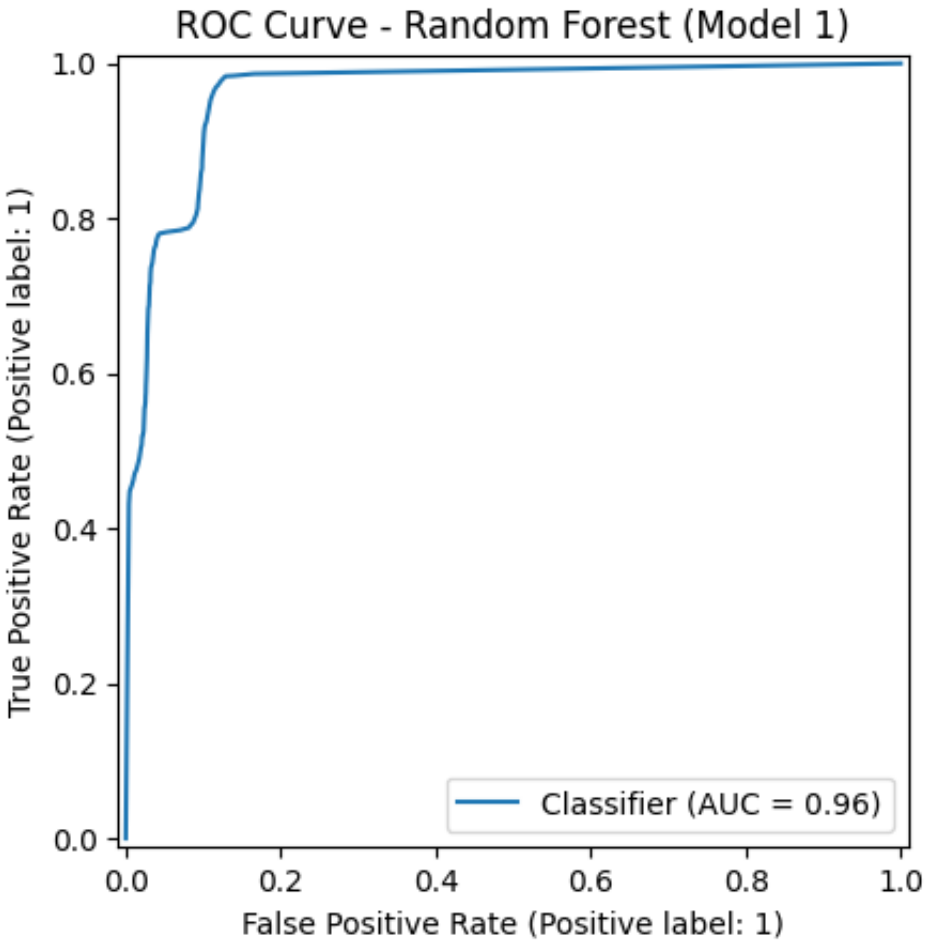
```
Random Forest trained
```

```
pred = rf.predict(X_test_sc)
prob = rf.predict_proba(X_test_sc)[:, 1]

acc = accuracy_score(y_test, pred)
roc = roc_auc_score(y_test, prob)
pr  = average_precision_score(y_test, prob)

print("Model 1 Results (Random Forest)")
print(f"Accuracy : {acc:.4f}")
print(f"ROC-AUC  : {roc:.4f}")
print(f"PR-AUC   : {pr:.4f}\n")

print(classification_report(y_test, pred, target_names=["Normal", "
```

```
Model 1 Results (Random Forest)
Accuracy : 0.7649
ROC-AUC  : 0.9619
PR-AUC   : 0.9645

              precision    recall  f1-score   support

      Normal       0.65      0.97      0.78      9711
      Attack       0.97      0.61      0.75     12833

    accuracy                           0.76     22544
   macro avg       0.81      0.79      0.76     22544
weighted avg       0.83      0.76      0.76     22544
```

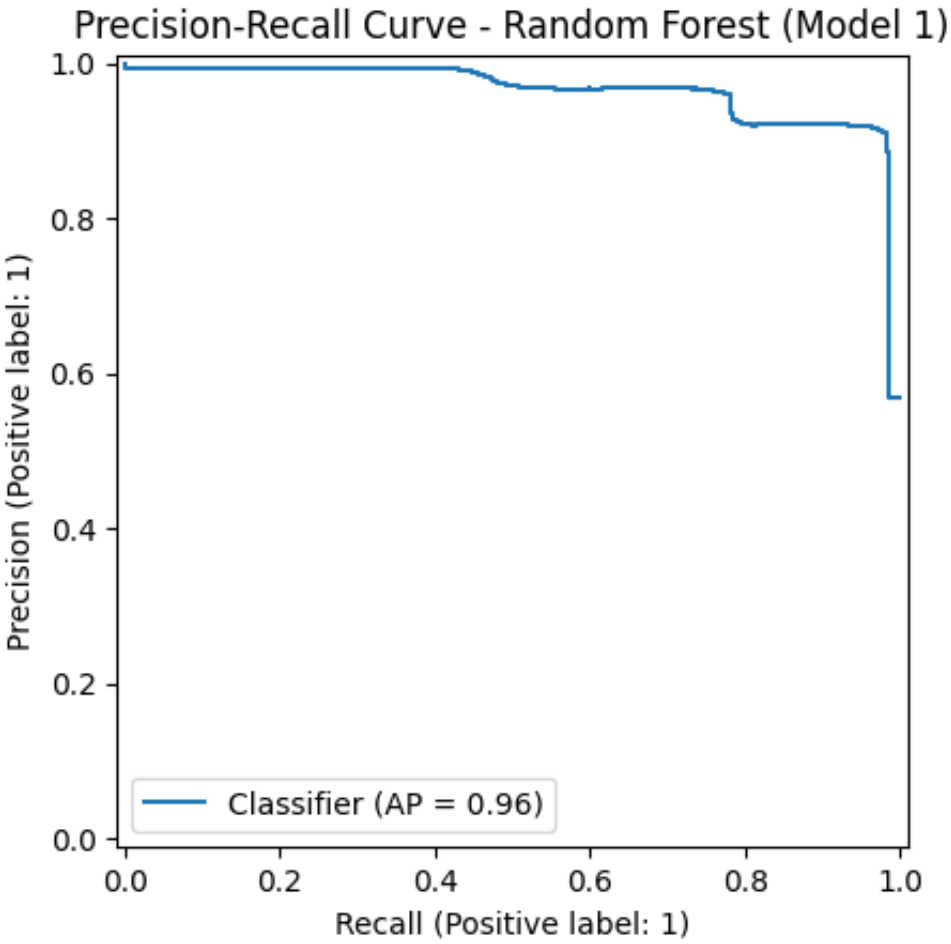
```
plt.figure(figsize=(6,5))
RocCurveDisplay.from_predictions(y_test, prob)
plt.title("ROC Curve - Random Forest (Model 1)")
plt.savefig("model1_rf_roc.png", dpi=300, bbox_inches="tight")
plt.show()
```

```
<Figure size 600x500 with 0 Axes>
```

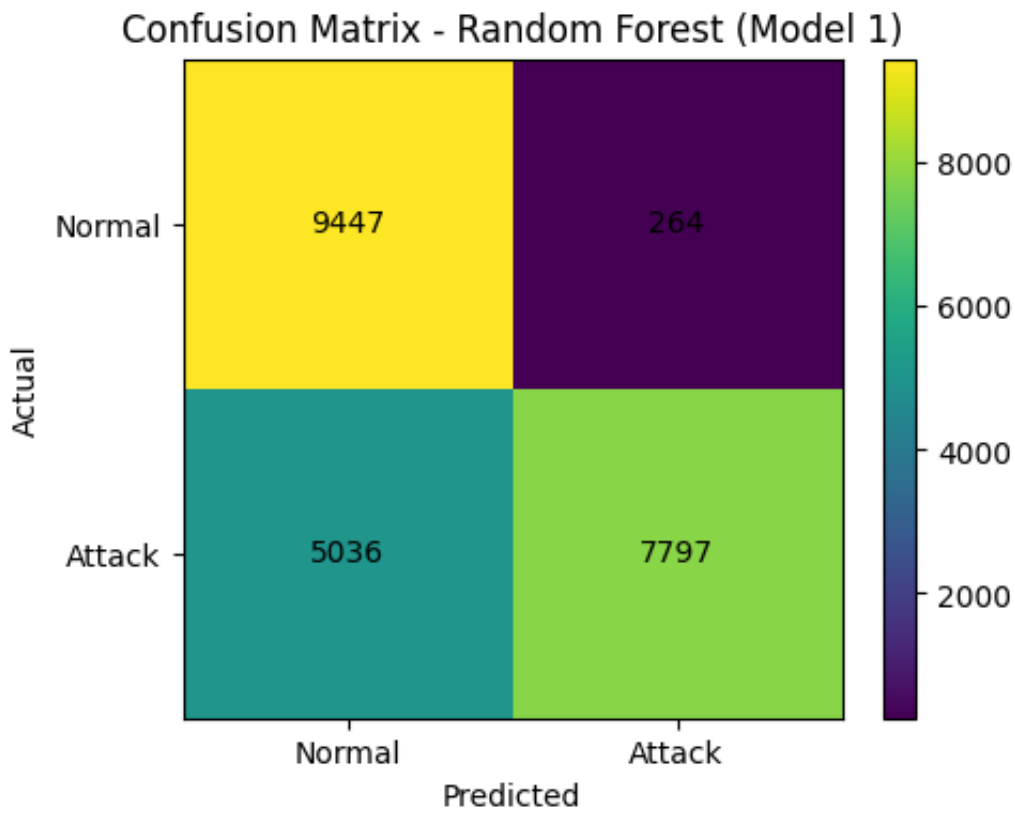


```
plt.figure(figsize=(6,5))
PrecisionRecallDisplay.from_predictions(y_test, prob)
plt.title("Precision-Recall Curve - Random Forest (Model 1)")
plt.savefig("model1_rf_pr.png", dpi=300, bbox_inches="tight")
plt.show()
```

```
<Figure size 600x500 with 0 Axes>
```

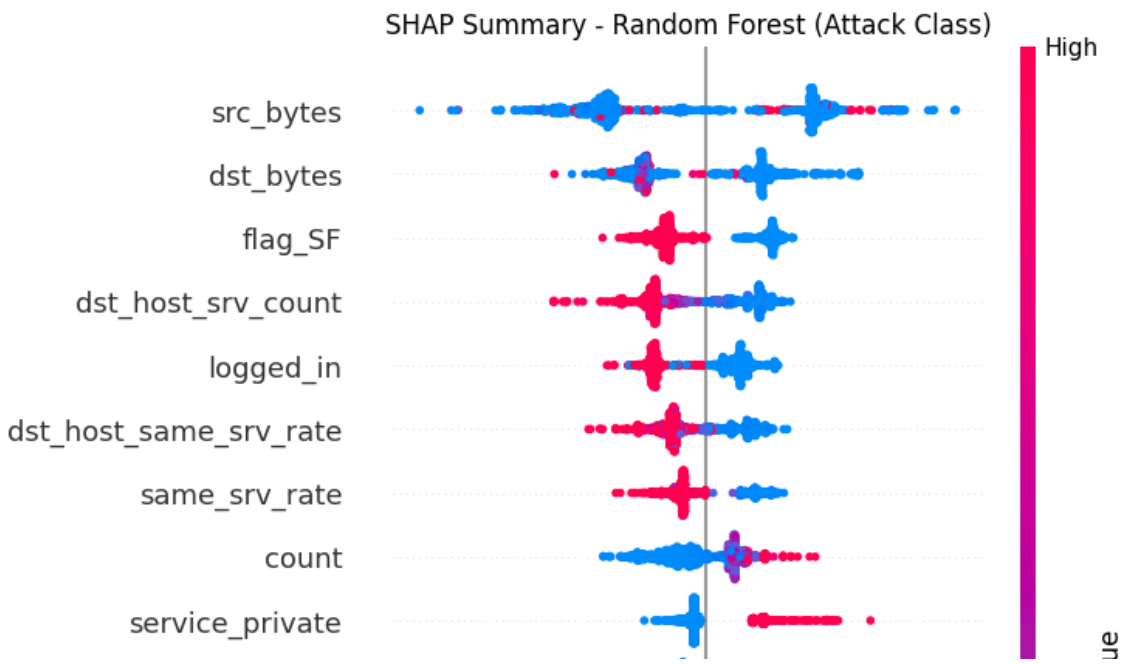


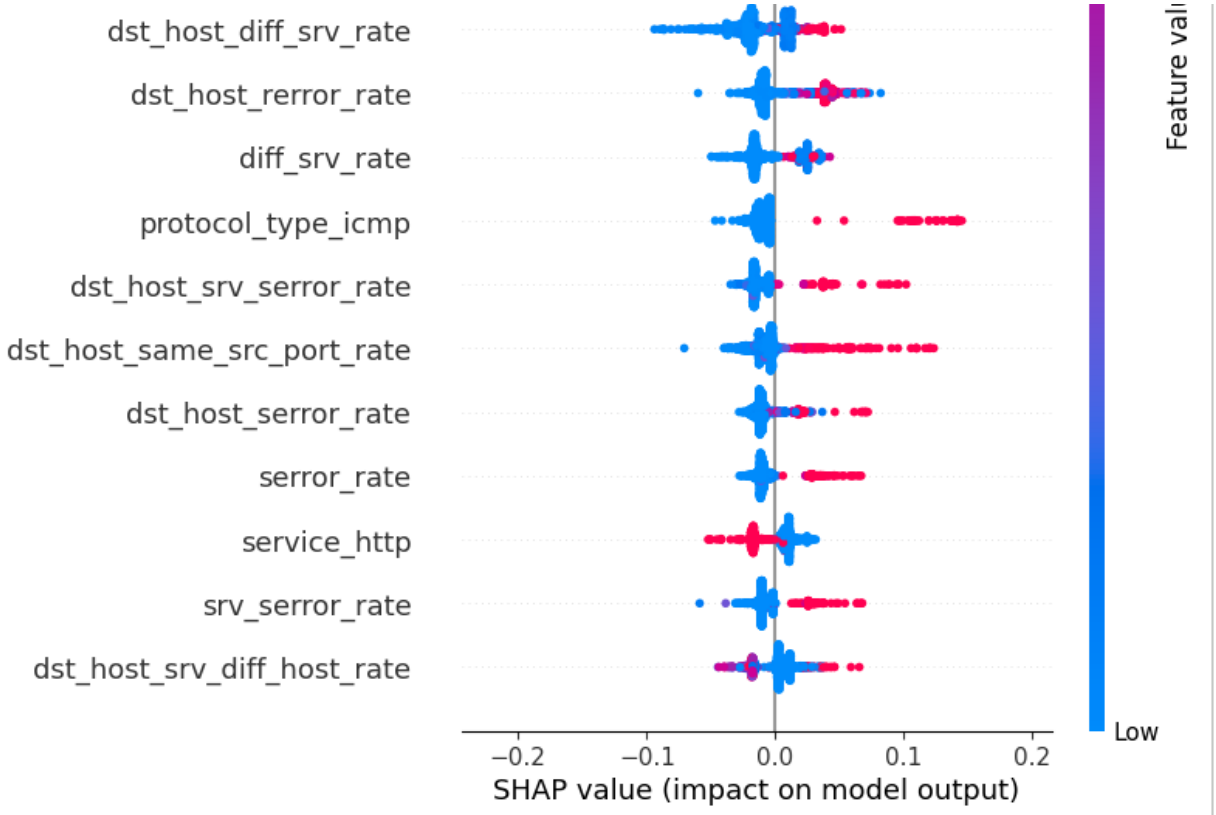
```
cm = confusion_matrix(y_test, pred)

plt.figure(figsize=(5,4))
plt.imshow(cm)
plt.title("Confusion Matrix - Random Forest (Model 1)")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.xticks([0,1], ["Normal","Attack"])
plt.yticks([0,1], ["Normal","Attack"])

for (i, j), v in np.ndenumerate(cm):
    plt.text(j, i, str(v), ha="center", va="center")

plt.colorbar()
plt.savefig("model1_rf_cm.png", dpi=300, bbox_inches="tight")
plt.show()
```



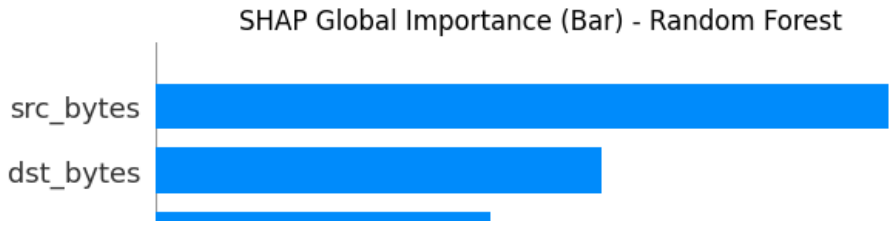

```
import shap
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# explain a subset
N = 1000
```

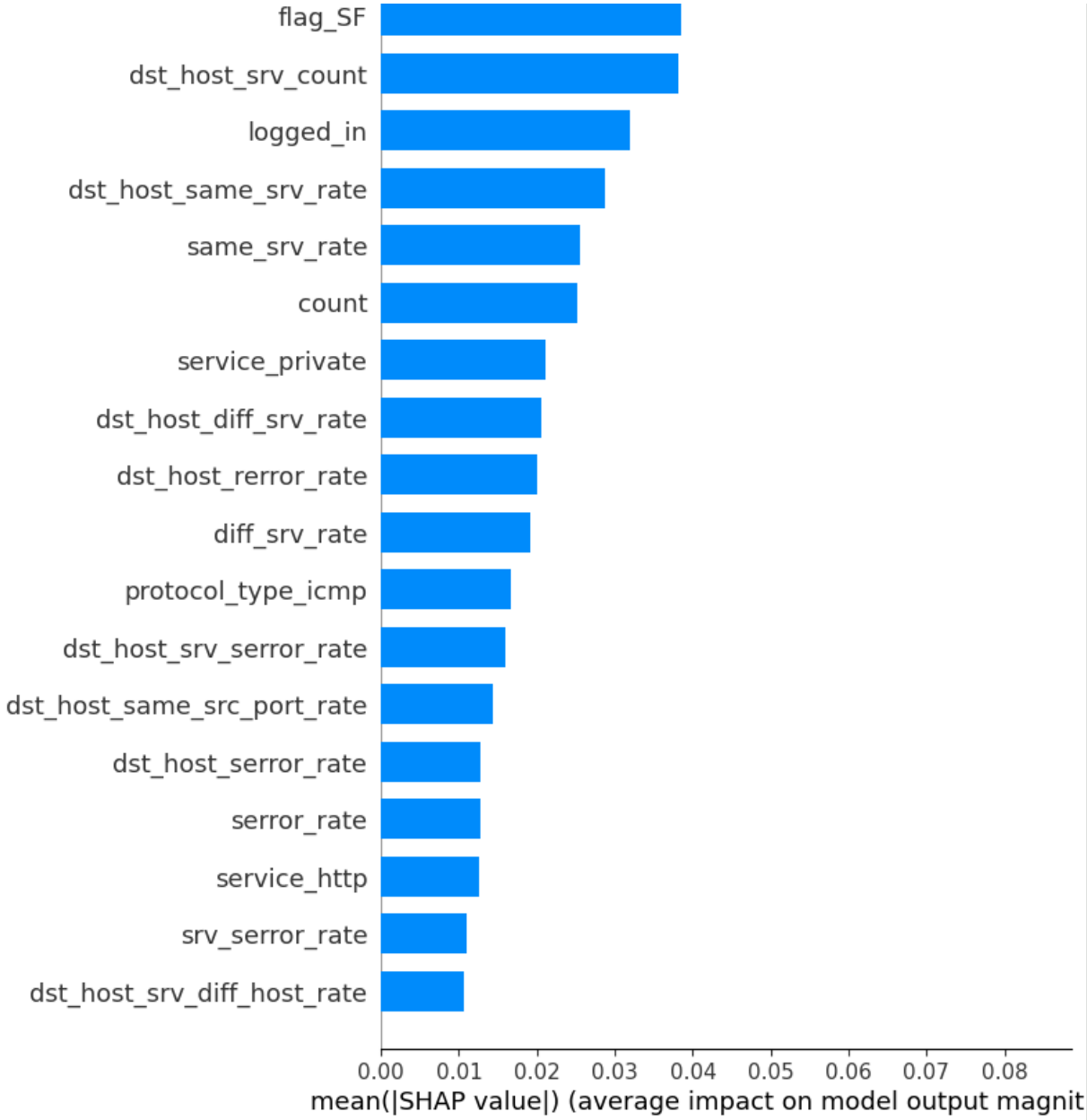
```
X_explain = X_test_sc[:N]


X_explain_df = pd.DataFrame(X_explain, columns=X_train.columns)

explainer = shap.TreeExplainer(rf)
shap_out = explainer.shap_values(X_explain_df)  # keep input consis

# Handle both SHAP output formats
# Case A: old format -> list [class0, class1]
if isinstance(shap_out, list):
    shap_attack = shap_out[1]  # attack class

# Case B: new format -> Explanation object (values: N x features x
# OR a 3D numpy array, which is the case here based on kernel state
else:
    # Directly check the ndim of shap_out (which is already a numpy
    if shap_out.ndim == 3:
        shap_attack = shap_out[:, :, 1]
    else:
        # fallback (rare): already N x features
        shap_attack = shap_out


# SHAP summary
plt.figure()
shap.summary_plot(shap_attack, X_explain_df, show=False)
plt.title("SHAP Summary - Random Forest (Attack Class)")
plt.savefig("model1_rf_shap_summary.png", dpi=300, bbox_inches="tig
plt.show()
```






```
# SHAP bar plot
plt.figure()
shap.summary_plot(shap_attack, X_explain_df, plot_type="bar", show=
plt.title("SHAP Global Importance (Bar) - Random Forest")
plt.savefig("model1_rf_shap_bar.png", dpi=300, bbox_inches="tight")
plt.show()
```




flag_SF
dst_host_srv_count
logged_in
dst_host_same_srv_rate
same_srv_rate
count
service_private
dst_host_diff_srv_rate
dst_host_rerror_rate
diff_srv_rate
protocol_type_icmp
dst_host_srv_serror_rate
dst_host_same_src_port_rate
dst_host_serror_rate
serror_rate
service_http
srv_serror_rate
dst_host_srv_diff_host_rate
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
mean(|SHAP value|) (average impact on model output magnit

```
# SHAP local explanation (one sample)
idx = 0
plt.figure()
shap.force_plot(
    explainer.expected_value[1] if isinstance(explainer.expected_va
    shap_attack[idx],
    X_explain_df.iloc[idx],
    matplotlib=True,
    show=False
)
plt.savefig("model1_rf_shap_local.png", dpi=300, bbox_inches="tight
plt.show()
```

```
<Figure size 640x480 with 0 Axes>
```

```
# These files let reuse Model 1 later without retraining, and they

joblib.dump(rf, "model1_random_forest.pkl")
joblib.dump(scaler, "scaler.pkl")
pd.Series(X_train.columns).to_csv("feature_names.csv", index=False)

print("Saved:")
print("- model1_random_forest.pkl")
print("- scaler.pkl")
print("- feature_names.csv")
```

```
Saved:
- model1_random_forest.pkl
- scaler.pkl
- feature_names.csv
```

```
import zipfile
import os

files_to_zip = [
    "model1_rf_roc.png",
    "model1_rf_pr.png",
    "model1_rf_cm.png",
    "model1_rf_shap_summary.png",
    "model1_rf_shap_bar.png",
    "model1_rf_shap_local.png",
    "model1_random_forest.pkl",
    "scaler.pkl",
    "feature_names.csv"
]

with zipfile.ZipFile("model1_outputs.zip", "w") as z:
    for f in files_to_zip:
        if os.path.exists(f):
            z.write(f)

print("Created: model1_outputs.zip")
```

```
Created: model1_outputs.zip
```

# Model 2 — Quantum Machine Learning Model (Variational Quantum Classifier)

This section implements an advanced quantum ML model using Qiskit. Because current quantum hardware and simulators are limited by qubit count, we reduce the feature space using PCA and map the reduced features into a quantum circuit. A Variational Quantum Classifier (VQC) is trained using a classical optimizer over a parameterized quantum circuit and evaluated using the same metrics as Model 1 to ensure a fair comparison.

```
!pip -q install qiskit qiskit-machine-learning qiskit-algorithms
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 8.9/8.9 MB 65.2 MB/s eta
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 263.1/263.1 kB 21.7 MB/s
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 327.8/327.8 kB 31.0 MB/s
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.2/2.2 MB 98.0 MB/s eta
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 54.4/54.4 kB 4.8 MB/s et
```

```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.decomposition import PCA
from sklearn.metrics import (
    accuracy_score, classification_report, confusion_matrix,
    roc_auc_score, average_precision_score,
    RocCurveDisplay, PrecisionRecallDisplay
)

# Qiskit import
from qiskit.circuit.library import ZZFeatureMap, RealAmplitudes

try:
    from qiskit.primitives import Sampler
except Exception:
    Sampler = None

try:
    from qiskit_algorithms.optimizers import COBYLA
except Exception:
    from qiskit.algorithms.optimizers import COBYLA

from qiskit_machine_learning.algorithms.classifiers import VQC
```

## The Variational Quantum Classifier combines a quantum feature map and trainable ansatz.

## Classical optimisation (COBYLA) updates circuit parameters to minimise classification loss.

```
# Model 2: Variational Quantum Classifier

from qiskit.circuit.library import ZZFeatureMap, RealAmplitudes


N_QUBITS = 6
TRAIN_SUBSET = 2000
TEST_SUBSET  = 1000

# PCA
pca = PCA(n_components=N_QUBITS, random_state=42)

X_train_q = pca.fit_transform(X_train_sc)
X_test_q  = pca.transform(X_test_sc)

# scale to [-pi, pi]
train_mins = X_train_q.min(axis=0)
train_maxs = X_train_q.max(axis=0)
denom = np.where((train_maxs - train_mins)==0, 1, (train_maxs-train

def to_pi(X):
    Xn = (X - train_mins)/denom
    return (Xn*2-1)*np.pi

X_train_q = to_pi(X_train_q)
X_test_q  = to_pi(X_test_q)

# subset
X_train_q_small = X_train_q[:TRAIN_SUBSET]
y_train_small   = np.array(y_train[:TRAIN_SUBSET])

X_test_q_small  = X_test_q[:TEST_SUBSET]
y_test_small    = np.array(y_test[:TEST_SUBSET])

print("Quantum shapes:", X_train_q_small.shape, X_test_q_small.shap

# quantum circuits
feature_map = ZZFeatureMap(feature_dimension=N_QUBITS, reps=1)
ansatz = RealAmplitudes(num_qubits=N_QUBITS, reps=1)

optimizer = COBYLA(maxiter=60)

if Sampler is not None:
    sampler = Sampler()
    vqc = VQC(feature_map=feature_map, ansatz=ansatz, optimizer=opt
else:
    vqc = VQC(feature_map=feature_map, ansatz=ansatz, optimizer=opt
```

```
print("Training quantum model...")
vqc.fit(X_train_q_small, y_train_small)
print("VQC trained successfully ")
```

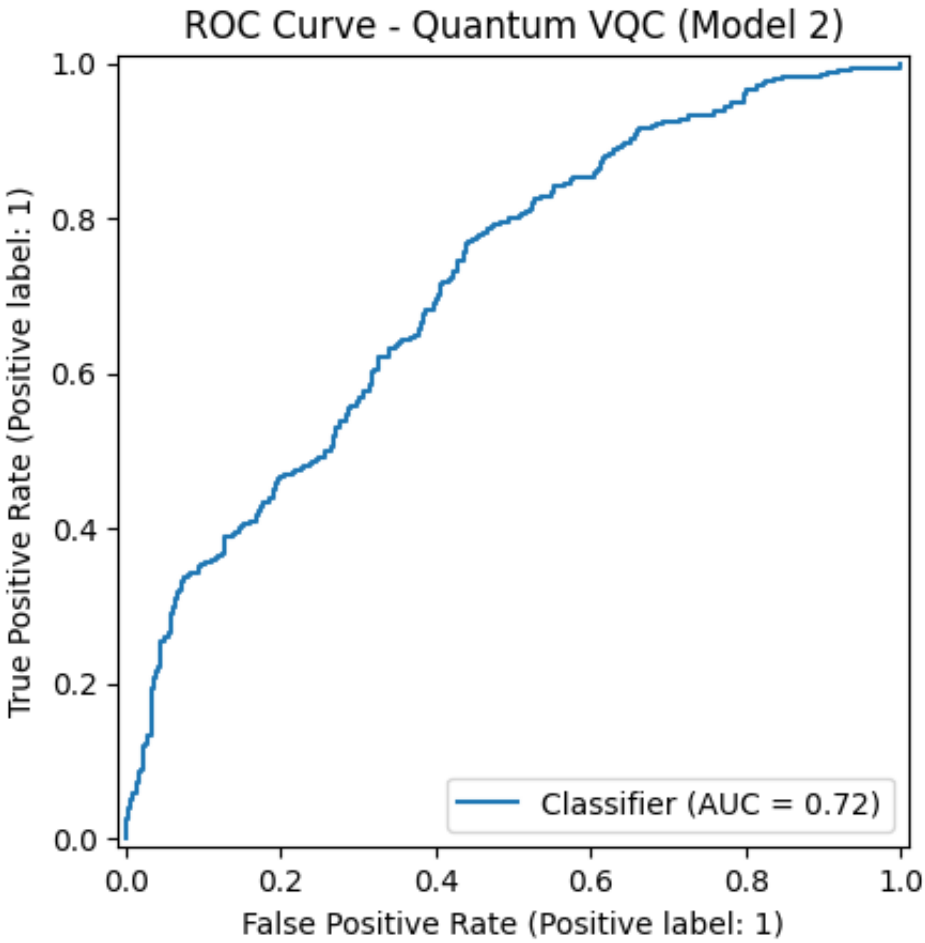
```
/tmp/ipython-input-832375762.py:38: DeprecationWarning: The class ``
  feature_map = ZZFeatureMap(feature_dimension=N_QUBITS, reps=1)
/tmp/ipython-input-832375762.py:39: DeprecationWarning: The class ``
  ansatz = RealAmplitudes(num_qubits=N_QUBITS, reps=1)
WARNING:qiskit_machine_learning.neural_networks.sampler_qnn:No gradi
Quantum shapes: (2000, 6) (1000, 6)
Training quantum model...
VQC trained successfully
```

```
y_pred_q  = vqc.predict(X_test_q_small)
y_score_q = vqc.predict_proba(X_test_q_small)[:, 1]

acc_q = accuracy_score(y_test_small, y_pred_q)
roc_q = roc_auc_score(y_test_small, y_score_q)
pr_q  = average_precision_score(y_test_small, y_score_q)

print("Model 2 Results (Quantum VQC)")
print(f"Accuracy : {acc_q:.4f}")
print(f"ROC-AUC  : {roc_q:.4f}")
print(f"PR-AUC   : {pr_q:.4f}\n")
print(classification_report(y_test_small, y_pred_q, target_names=["
```

```
Model 2 Results (Quantum VQC)
Accuracy : 0.6480
ROC-AUC  : 0.7156
PR-AUC   : 0.7471

              precision    recall  f1-score   support

      Normal       0.61      0.60      0.61       452
      Attack       0.68      0.69      0.68       548

    accuracy                           0.65      1000
   macro avg       0.64      0.64      0.64      1000
weighted avg       0.65      0.65      0.65      1000
```

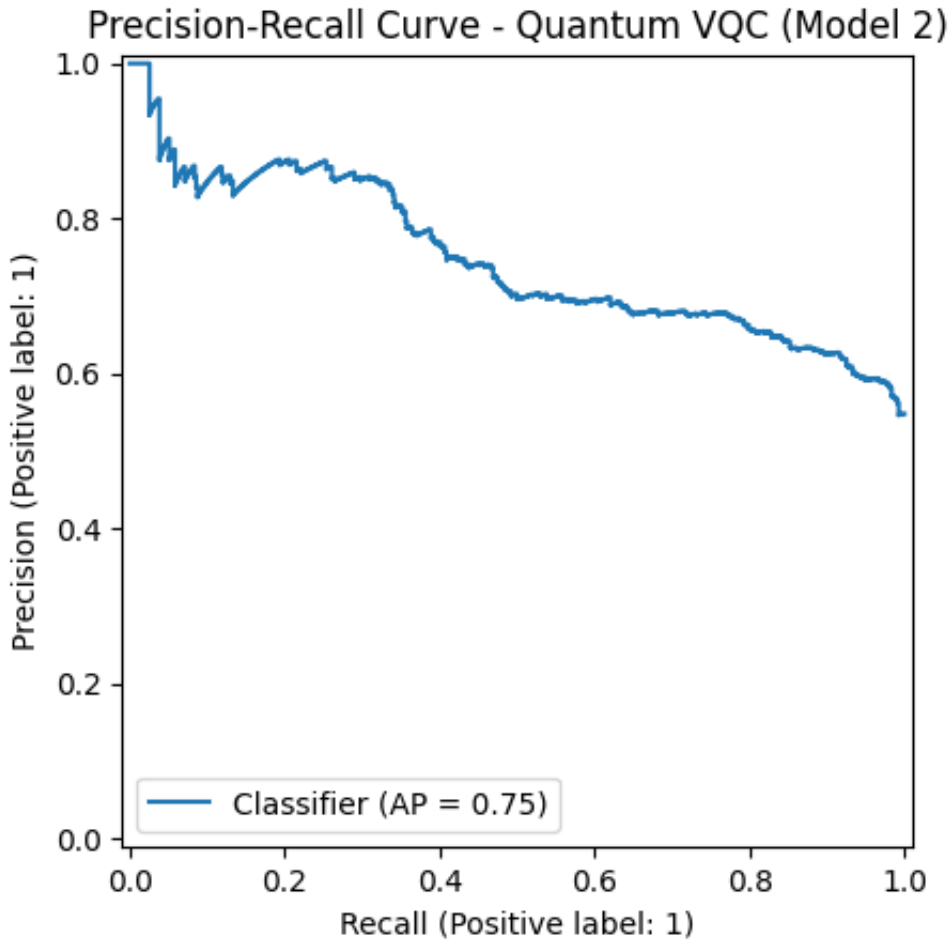
```
plt.figure(figsize=(6,5))
RocCurveDisplay.from_predictions(y_test_small, y_score_q)
plt.title("ROC Curve - Quantum VQC (Model 2)")
plt.savefig("model2_vqc_roc.png", dpi=300, bbox_inches="tight")
plt.show()
```

```
<Figure size 600x500 with 0 Axes>
```

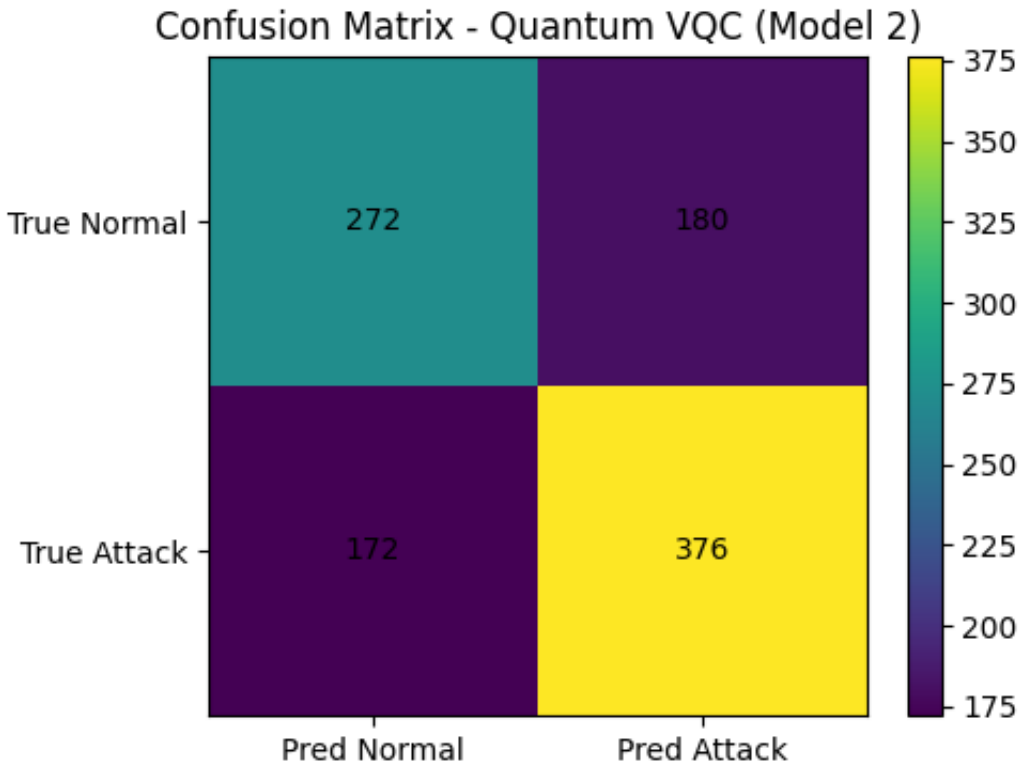


```
plt.figure(figsize=(6,5))
PrecisionRecallDisplay.from_predictions(y_test_small, y_score_q)
plt.title("Precision-Recall Curve - Quantum VQC (Model 2)")
plt.savefig("model2_vqc_pr.png", dpi=300, bbox_inches="tight")
plt.show()
```

```
<Figure size 600x500 with 0 Axes>
```

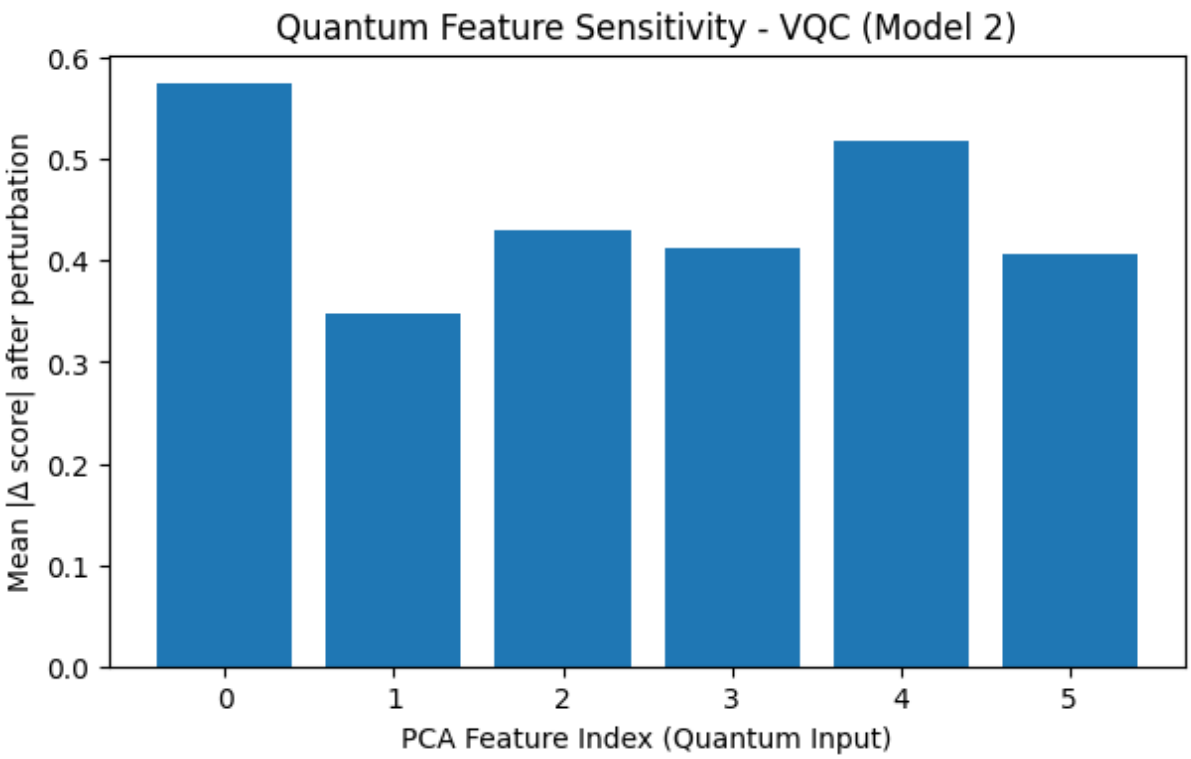


```
cm_q = confusion_matrix(y_test_small, y_pred_q)

plt.figure(figsize=(5,4))
plt.imshow(cm_q)
plt.title("Confusion Matrix - Quantum VQC (Model 2)")
plt.xticks([0,1], ["Pred Normal","Pred Attack"])
plt.yticks([0,1], ["True Normal","True Attack"])

for (i, j), v in np.ndenumerate(cm_q):
    plt.text(j, i, str(v), ha="center", va="center")

plt.colorbar()
plt.savefig("model2_vqc_cm.png", dpi=300, bbox_inches="tight")
plt.show()
```




```
# Quantum XAI: Sensitivity Analysis
# Measure how much the model score changes when each PCA feature is

def get_scores(model, X):
    if hasattr(model, "decision_function"):
        s = model.decision_function(X)
        return 1 / (1 + np.exp(-s))
    return model.predict(X).astype(float)

base_scores = get_scores(vqc, X_test_q_small)
```

```
eps = 0.15  # small perturbation (radians)
importances = []

for j in range(N_QUBITS):
    Xp = X_test_q_small.copy()
    Xp[:, j] += eps
    pert_scores = get_scores(vqc, Xp)
    importances.append(np.mean(np.abs(pert_scores - base_scores)))

importances = np.array(importances)

plt.figure(figsize=(7,4))
plt.bar(range(N_QUBITS), importances)
plt.title("Quantum Feature Sensitivity - VQC (Model 2)")
plt.xlabel("PCA Feature Index (Quantum Input)")
plt.ylabel("Mean |Δ score| after perturbation")
plt.savefig("model2_vqc_sensitivity.png", dpi=300, bbox_inches="tigh
plt.show()

print("Saved: model2_vqc_sensitivity.png")
```




```
Saved: model2_vqc_sensitivity.png
```

```
# Save all Model 2 output figures into one zip file

import zipfile, os

files = [
    "model2_vqc_roc.png",
    "model2_vqc_pr.png",
    "model2_vqc_cm.png",
    "model2_vqc_sensitivity.png"
]

zip_name = "model2_outputs.zip"

with zipfile.ZipFile(zip_name, "w") as z:
    for f in files:
        if os.path.exists(f):
            z.write(f)

print(f"{zip_name} created successfully")
```

```
model2_outputs.zip created successfully
```

# Model 3 — Hybrid Quantum–Classical Model

This section combines predictions from the classical Random Forest model and the quantum VQC model to create a hybrid anomaly detection system.

The goal is to leverage the stability of classical machine learning and the feature representation capability of quantum models. The combined predictions are used to train a meta-classifier to improve detection performance.

```
# Classical model probabilities (from Random Forest)
rf_probs = rf.predict_proba(X_test_sc)[:, 1]

# use same subset size as quantum test
rf_probs_small = rf_probs[:len(y_test_small)]
```

```
q_probs_small = y_score_q
```

```
import numpy as np

# combine classical + quantum predictions
X_hybrid = np.column_stack((rf_probs_small, q_probs_small))
y_hybrid = y_test_small

print("Hybrid feature shape:", X_hybrid.shape)
```

```
Hybrid feature shape: (1000, 2)
```

```
from sklearn.linear_model import LogisticRegression

hybrid_model = LogisticRegression()
hybrid_model.fit(X_hybrid, y_hybrid)

print("Hybrid model trained ")
```

```
Hybrid model trained
```

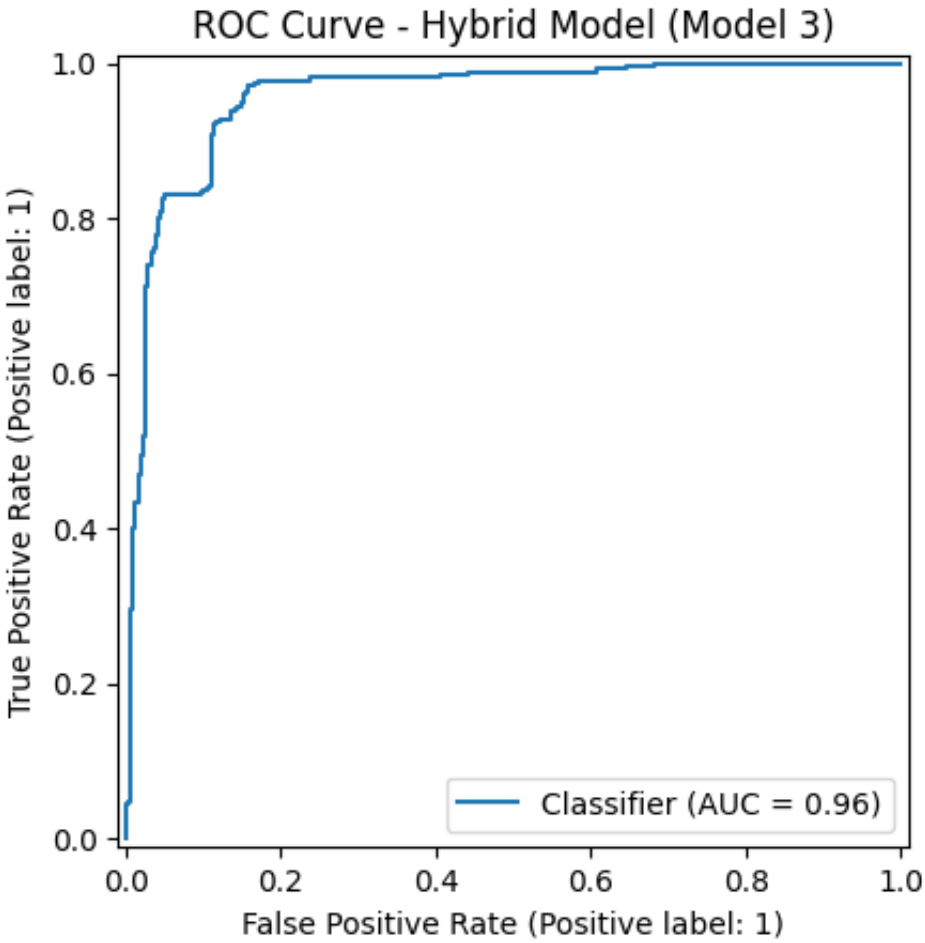
```
hybrid_pred = hybrid_model.predict(X_hybrid)
hybrid_prob = hybrid_model.predict_proba(X_hybrid)[:,1]
```

```
from sklearn.metrics import accuracy_score, roc_auc_score, average_
from sklearn.metrics import classification_report, confusion_matrix

acc_h = accuracy_score(y_hybrid, hybrid_pred)
roc_h = roc_auc_score(y_hybrid, hybrid_prob)
pr_h  = average_precision_score(y_hybrid, hybrid_prob)

print("Hybrid Model Results")
print(f"Accuracy : {acc_h:.4f}")
print(f"ROC-AUC  : {roc_h:.4f}")
print(f"PR-AUC   : {pr_h:.4f}\n")

print(classification_report(y_hybrid, hybrid_pred, target_names=["N
```

```
Hybrid Model Results
Accuracy : 0.8690
ROC-AUC  : 0.9568
PR-AUC   : 0.9546

              precision    recall  f1-score   support

      Normal       0.79      0.96      0.87       452
      Attack       0.96      0.80      0.87       548

    accuracy                           0.87      1000
   macro avg       0.88      0.88      0.87      1000
weighted avg       0.88      0.87      0.87      1000
```

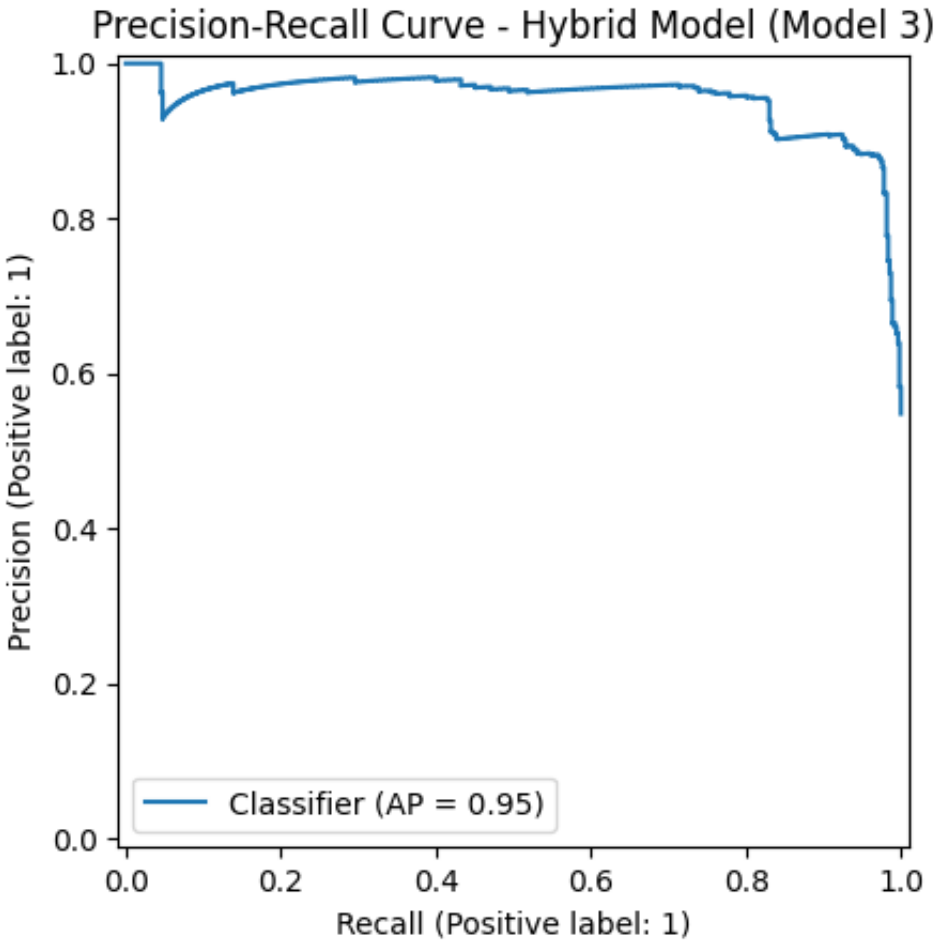
```
plt.figure(figsize=(6,5))
RocCurveDisplay.from_predictions(y_hybrid, hybrid_prob)
plt.title("ROC Curve - Hybrid Model (Model 3)")
plt.savefig("model3_hybrid_roc.png", dpi=300, bbox_inches="tight")
plt.show()
```

```
<Figure size 600x500 with 0 Axes>
```

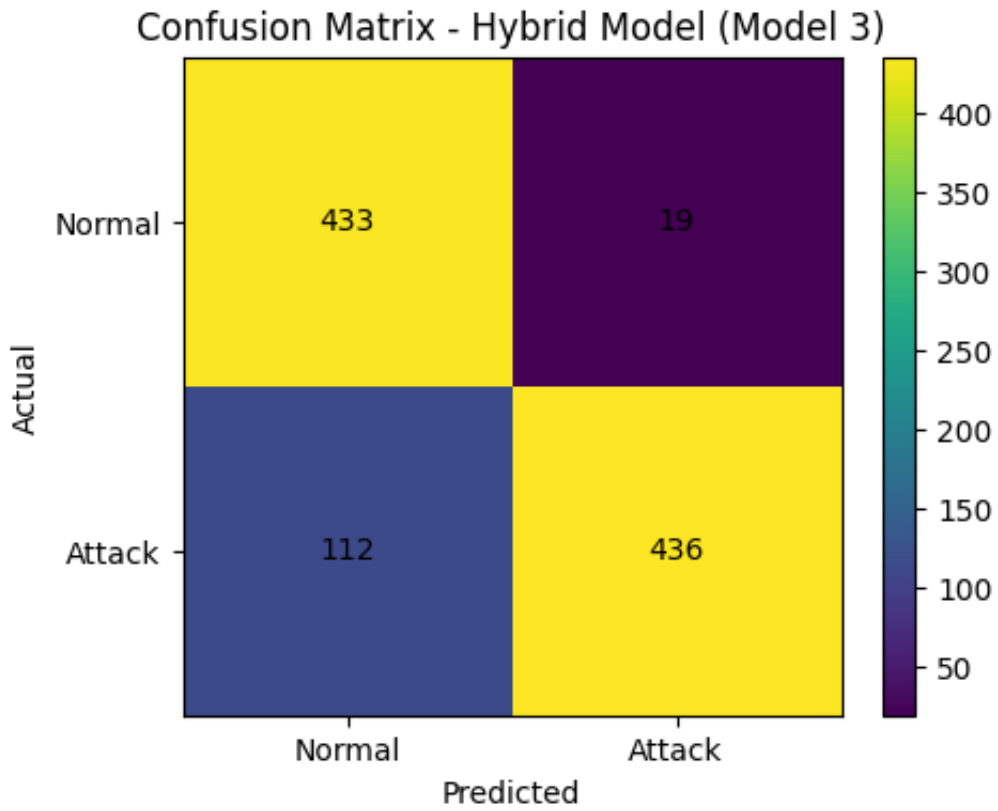


```
plt.figure(figsize=(6,5))
PrecisionRecallDisplay.from_predictions(y_hybrid, hybrid_prob)
plt.title("Precision-Recall Curve - Hybrid Model (Model 3)")
plt.savefig("model3_hybrid_pr.png", dpi=300, bbox_inches="tight")
plt.show()
```

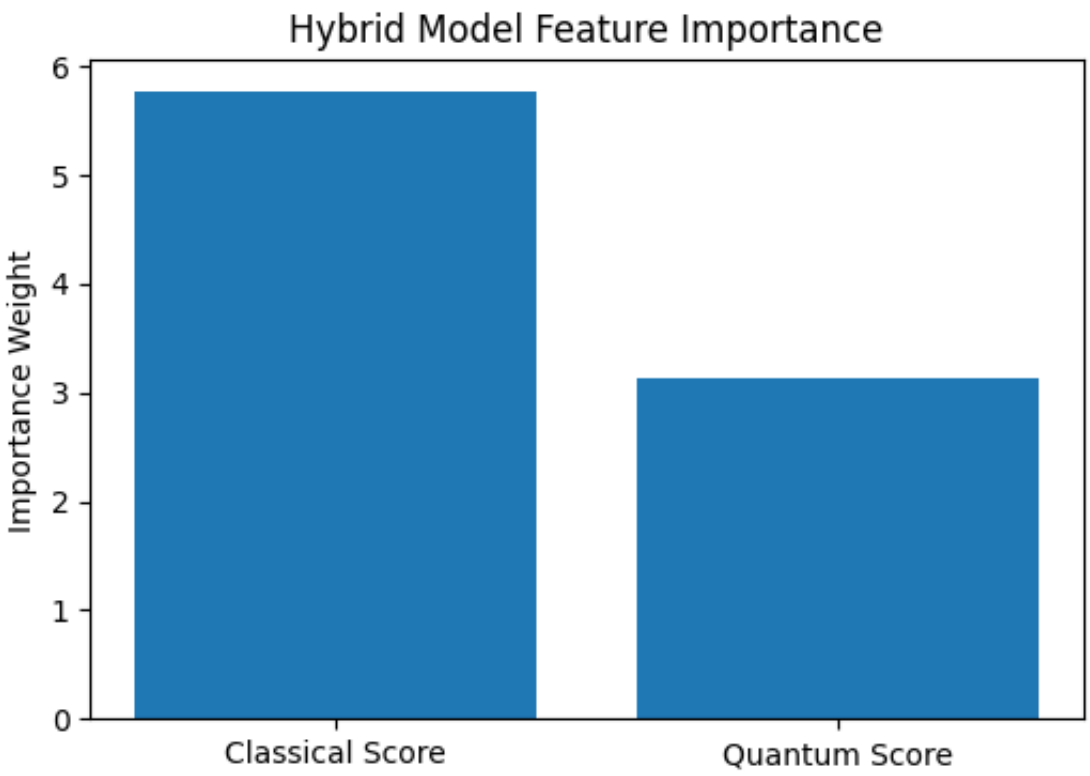
```
<Figure size 600x500 with 0 Axes>
```



```
cm_h = confusion_matrix(y_hybrid, hybrid_pred)

plt.figure(figsize=(5,4))
plt.imshow(cm_h)
plt.title("Confusion Matrix - Hybrid Model (Model 3)")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.xticks([0,1], ["Normal","Attack"])
plt.yticks([0,1], ["Normal","Attack"])

for (i, j), v in np.ndenumerate(cm_h):
    plt.text(j, i, str(v), ha="center", va="center")

plt.colorbar()
plt.savefig("model3_hybrid_cm.png", dpi=300, bbox_inches="tight")
plt.show()
```



```
# Hybrid Model XAI
# Examine how much classical vs quantum predictions
# influence the hybrid model decisions.

import matplotlib.pyplot as plt

importance = hybrid_model.coef_[0]
features = ["Classical Score", "Quantum Score"]

plt.figure(figsize=(6,4))
plt.bar(features, importance)
plt.title("Hybrid Model Feature Importance")
plt.ylabel("Importance Weight")
plt.savefig("model3_hybrid_xai.png", dpi=300, bbox_inches="tight")
plt.show()
```



```
import zipfile, os

files = [
    "model3_hybrid_roc.png",
    "model3_hybrid_pr.png",
    "model3_hybrid_cm.png",
    "model3_hybrid_xai.png"
]

with zipfile.ZipFile("model3_outputs.zip", "w") as z:
    for f in files:
        if os.path.exists(f):
            z.write(f)

print("model3_outputs.zip created")
```

```
model3_outputs.zip created
```

# Conclusion

Three models were implemented and evaluated for anomaly detection: a classical Random Forest, a quantum VQC model, and a hybrid approach.

The hybrid model achieved the best overall performance, improving attack detection while maintaining strong ROC and PR performance.

These results demonstrate the potential of hybrid quantum–classical approaches for future anomaly detection systems.

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.